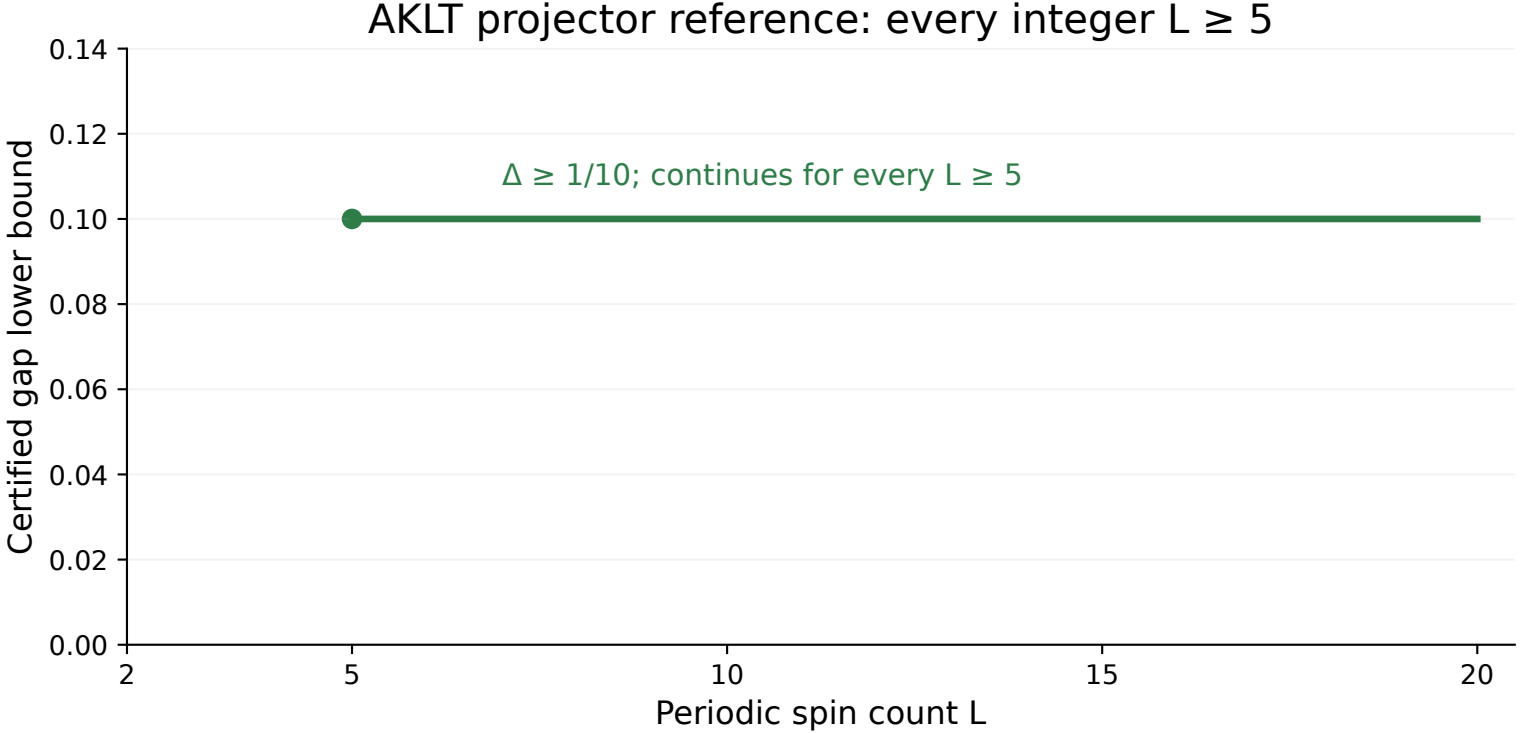

H = ΣP₂; unique ground for L ≥ 3. D = 2 MPS. Separate finite-size projector proof, not thermal bootstrap.